

Leaf Disease Detection Using Convolutional Neural Network

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DECLARATION OF ORIGINALITY

We hereby declare that this Final Year Project Report entitled “Plant Leaf Disease Detection Using CNN” is our original work carried out under the guidance of Dr. Zeenat Rehena. This project has not been submitted earlier, either in part or in full, for the award of any degree or diploma to any other University or Institution.

All sources of information used in this report have been duly acknowledged.

Abstract

Agriculture plays a vital role in the economy and has a significant impact on human life. Crop productivity is highly affected by plant leaf diseases, and early detection of such diseases is essential to reduce economic losses. However, manual identification of plant diseases requires expert knowledge and is time-consuming, making it difficult for farmers to monitor large agricultural fields efficiently.

This project focuses on Plant Leaf Disease Detection using Convolutional Neural Networks (CNN) to automatically identify and classify diseases from leaf images. The dataset used in this work is divided into training, validation, and testing sets to ensure effective learning and reliable performance evaluation. Image preprocessing techniques are applied, and a deep CNN model is designed using multiple convolutional, pooling, and dense layers to extract discriminative features from leaf images.

The proposed model is capable of detecting disease symptoms at an early stage, which helps in timely treatment and improved crop yield. This automated approach reduces dependency on manual inspection and provides a scalable solution for monitoring large agricultural fields. The system has wide applications in smart agriculture, biological research, and agricultural institutions, making plant leaf disease detection an important and practical research area.

Acknowledgments

This project report is based on the work carried out for **“Plant Leaf Disease Detection Using Convolutional Neural Network”**. The successful completion of this project would not have been possible without the guidance and support of several individuals.

I consider it a privilege to work under the guidance of my project supervisor, **Dr. Zeenat Rehena**, whose valuable suggestions, constant encouragement, and technical guidance greatly contributed to the completion of this project. Her support and timely feedback helped me throughout the development and implementation of the work.

I am grateful to Aliah University for providing the necessary facilities, resources, and academic environment to carry out this project successfully. I also extend my sincere thanks to all the faculty members and friends who directly or indirectly supported me during the course of this work.

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Chapter 1: Introduction

1.1 Introduction of Leaf Disease Detection

Agriculture is one of the most important sectors of the economy, and plant diseases have a significant impact on crop yield and quality. Leaf diseases are among the primary causes of reduced agricultural production. Early and accurate detection of leaf diseases is essential to minimize crop loss and improve overall productivity. However, farmers often lack expert knowledge to identify plant diseases at an early stage, which leads to delayed treatment and economic loss.

To overcome this challenge, deep learning–based plant disease detection techniques have gained considerable attention in recent years. In this work, a **Convolutional Neural Network (CNN)**–based approach is proposed to automatically detect and classify plant leaf diseases using image data.

The dataset used in this study consists of **RGB images of healthy and diseased plant leaves**, covering **38 different classes**. These classes include multiple crops such as **apple, grape, corn, potato, and tomato**, along with their respective disease categories and healthy leaf samples.

The dataset is organized while preserving the directory structure and is divided as follows:

- **Training set:** 43,456 images belonging to 38 classes
- **Validation set:** 10,849 images belonging to 38 classes

All images are preprocessed and resized to a fixed resolution before being used for model training. The diseases present in the dataset include, but are not limited to, apple scab, black rot, grape leaf blight, corn rust, potato early and late blight, and tomato bacterial spot, leaf mold, and mosaic virus.



1. Apple scab 2. Grape Esca 3. Corn leaf spot 4. potato Early Blight 5. Tomato Bacterial Spot

Fig.1 Leaves with Disease part

Figure 1 illustrates sample images of healthy and diseased leaves from different crops such as potato, tomato, corn, apple, and grape. The visible disease patterns in the leaves make them suitable for detection using CNN-based deep learning models.

The proposed CNN model is trained on the augmented dataset, and its performance is evaluated using the validation set to analyze classification accuracy and efficiency.

1.2 Applications

- Biological research
- Plant leaf disease detection also useful in agriculture institute.
- Some plant leaf disease detection automatic techniques are beneficial for large work of monitoring in farm of crops disease detection.

1.3 Objectives

- The objective of this research is to concentrate based on potato, tomato, corn, grapes, and apple leaf disease detection using CNN.
- CNN is used for examine the healthy and diseased plants leaves.

1.4 Motivation

- Identifying and recognition of leaves disease is the solution for saving the reduction of large farm in crop disease detection and profit in productivity, it is beneficial in agricultural institute, Biological research.

Chapter 2: Existing Work & Implementation work

2.1 Overview of Existing Work

Existing research on leaf disease detection has shown that Convolutional Neural Networks (CNNs) are highly effective for automatically detecting and classifying plant leaf diseases using image data. Earlier approaches relied heavily on traditional image processing and hand-crafted feature extraction techniques. However, recent studies focus on deep learning–based models that learn discriminative features directly from images, reducing manual intervention and improving accuracy.

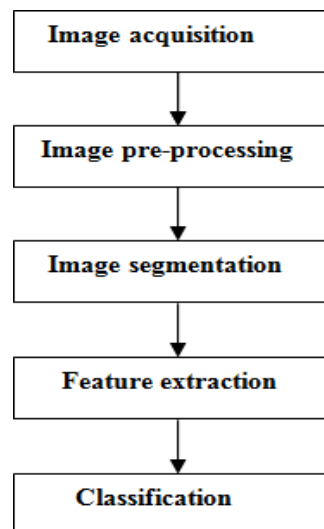


Figure :- 2

Figure 2 shows the **general block diagram of a CNN-based leaf disease detection approach**, which typically consists of the following stages:

A. Image Acquisition:

Image acquisition is the first step, where leaf images are captured using digital cameras or collected from publicly available datasets. These images are stored in digital format and used as input for further processing and model training.

B. Image Preprocessing:

Image preprocessing is performed to improve image quality and make the data suitable for CNN models. This step includes resizing images to a fixed dimension, normalization of pixel values, noise removal, and data augmentation techniques such as rotation, flipping, and scaling. These techniques help improve model generalization and reduce overfitting.

C. Feature Learning using CNN:

Unlike traditional methods that use manual feature extraction techniques such as GLCM, CNN-based approaches automatically learn relevant features from raw images through convolutional, pooling, and activation layers. These layers capture important patterns such as edges, textures, shapes, and disease-specific visual symptoms present on leaf surfaces.

D. Classification:

The learned features are passed through fully connected layers followed by a softmax classifier to categorize the input image into healthy or diseased classes. CNN models provide high accuracy due to their ability to learn complex, non-linear representations directly from image data.

In recent studies, CNN-based models have outperformed traditional machine learning approaches such as Random Forest and Support Vector Machines by eliminating the need for explicit segmentation and manual feature extraction, making them more robust and scalable for large agricultural datasets.

2.2 Implementation work

In this work, a **Convolutional Neural Network (CNN)**–based leaf disease detection system is implemented using a large-scale image dataset of plant leaves. The dataset consists of **RGB images of healthy and diseased crop leaves**, categorized into **38 different classes**. The crops included in the dataset are **apple, grape, corn, potato, and tomato**, each containing multiple disease categories along with healthy leaf samples.

The dataset is organized into subsets while preserving the directory structure:

- **Training set:** 43,456 images belonging to 38 classes
- **Validation set:** 10,849 images belonging to 38 classes
- **Test set:** A small set of leaf images used separately for final prediction and performance testing

All images are preprocessed and resized to a fixed resolution before being fed into the CNN model. Data augmentation techniques are applied to improve model generalization and reduce overfitting. The trained CNN model effectively learns discriminative features from the leaf images to accurately classify different plant diseases.

All images are resized to a fixed input size suitable for the CNN model and normalized to improve training stability. **Image preprocessing and data augmentation** techniques such as rotation, flipping, scaling, and zooming are applied to increase data diversity and reduce overfitting.

Proposed Workflow

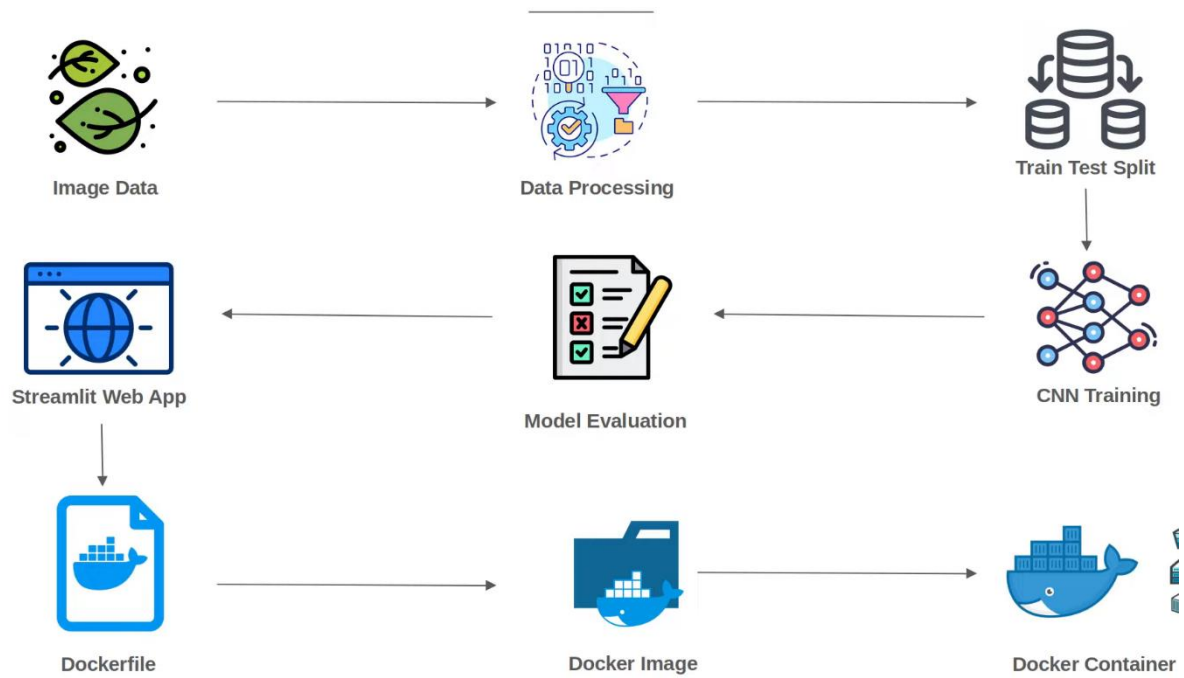


Figure :- 3

Figure 3 illustrates the proposed workflow of the CNN-based leaf disease detection system. The workflow begins with the input of leaf images, followed by preprocessing and augmentation steps. The dataset is then divided into training and validation sets. The CNN model is trained using the training dataset, while the validation dataset is used to evaluate model performance during training and apply early stopping to prevent overfitting. After training, the optimized model is used for predicting disease classes from new leaf images.

CNN Model Architecture

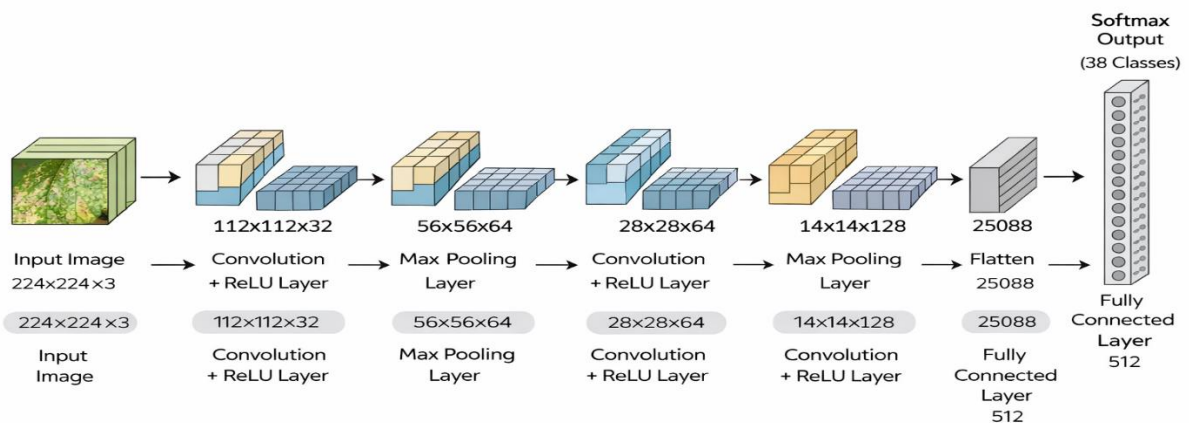


Figure :- 4

Figure 4 shows the architecture of the proposed Convolutional Neural Network used for leaf disease detection. The CNN architecture is designed to automatically learn discriminative features from leaf images without manual feature extraction.

The model consists of multiple **convolutional layers** followed by **ReLU activation functions** and **max-pooling layers**. These layers extract low-level and high-level features such as edges, texture patterns, color variations, and disease-specific visual symptoms present on leaf surfaces. As the depth of the network increases, the complexity of the learned features also increases.

Convolutional layers apply learnable filters to extract spatial features from the input images. Max-pooling layers reduce the spatial dimensions of feature maps, which decreases computational complexity and improves model robustness. ReLU activation functions introduce non-linearity into the network, enabling the CNN to learn complex representations.

To reduce overfitting, **dropout regularization** is applied during training. The extracted feature maps are flattened and passed through **fully connected layers**, followed by a **softmax output layer**, which classifies the input image into one of the **38 predefined leaf disease classes**.

Model Training and Evaluation

The CNN model is trained using the training dataset, and its performance is continuously monitored using the validation dataset. **Early stopping** is applied based on validation loss and accuracy to ensure optimal model performance and prevent overfitting. Model accuracy is computed at the end of each training epoch and is used as the primary performance evaluation metric.

Prediction Phase

After training, the final CNN model is used for prediction. A user can provide a **new leaf image** as input to the system. The trained CNN processes the image and outputs the predicted class label, indicating whether the leaf is **healthy or diseased**, along with the corresponding disease type and a confidence score representing prediction reliability.

Chapter 3: Dataset, Implementation and Result

3.1 Dataset Detail

The dataset used in this work is obtained from **Kaggle** and is a widely used **Plant Leaf Disease dataset** for agricultural image classification tasks. The dataset consists of approximately **87,000 RGB images** of healthy and diseased crop leaves, making it suitable for training deep learning models.

The dataset contains **38 different classes**, representing multiple disease categories and healthy leaf samples across five crops: **apple, grape, corn, potato, and tomato**. Each class corresponds to a specific crop–disease combination or a healthy leaf condition.

```
{0: 'Apple__Apple_scab',
 1: 'Apple__Black_rot',
 2: 'Apple__Cedar_apple_rust',
 3: 'Apple__healthy',
 4: 'Blueberry__healthy',
 5: 'Cherry_(including_sour)__Powdery_mildew',
 6: 'Cherry_(including_sour)__healthy',
 7: 'Corn_(maize)__Cercospora_leaf_spot Gray_leaf_spot',
 8: 'Corn_(maize)__Common_rust_',
 9: 'Corn_(maize)__Northern_Leaf_Blight',
10: 'Corn_(maize)__healthy',
11: 'Grape__Black_rot',
12: 'Grape__Esca_(Black_Measles)',
13: 'Grape__Leaf_blight_(Isariopsis_Leaf_Spot)',
14: 'Grape__healthy',
15: 'Orange__Haunglongbing_(Citrus_greening)',
16: 'Peach__Bacterial_spot',
17: 'Peach__healthy',
18: 'Pepper,_bell__Bacterial_spot',
19: 'Pepper,_bell__healthy',
20: 'Potato__Early_blight',
21: 'Potato__Late_blight',
22: 'Potato__healthy',
23: 'Raspberry__healthy',
24: 'Soybean__healthy',
...
33: 'Tomato__Spider_mites Two-spotted_spider_mite',
34: 'Tomato__Target_Spot',
35: 'Tomato__Tomato_Yellow_Leaf_Curl_Virus',
36: 'Tomato__Tomato_mosaic_virus',
37: 'Tomato__healthy'}
```

Dataset Organization

The dataset is organized into three directories while preserving the original folder structure:

- **Training set:** 43,456 images belonging to 38 classes
- **Validation set:** 10,849 images belonging to 38 classes
- **Test set:** A small set of leaf images used separately for final prediction and performance testing

The training and validation sets are used for learning and performance evaluation, while the test set is created separately for final disease prediction.

Image Properties

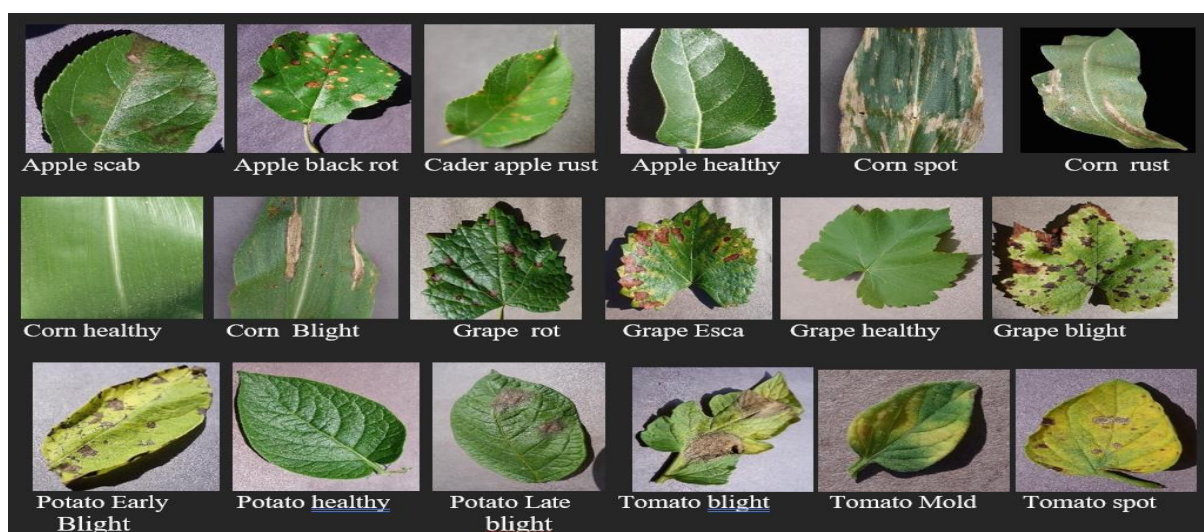
All images in the dataset are **RGB color images** captured under varying conditions such as different lighting, backgrounds, and leaf orientations. Before training, the images are resized to a fixed input size compatible with the CNN model and normalized to ensure stable and efficient training.

Data Augmentation

To enhance model performance and reduce overfitting, **offline and online data augmentation techniques** are applied. These include image rotation, flipping, scaling, and zooming. Data augmentation increases dataset diversity and helps the CNN model generalize better to unseen leaf images.

Class Distribution

The dataset includes both **healthy and diseased leaf images** for each crop. Disease categories include common plant diseases such as blight, rust, leaf spot, mold, and viral infections. The presence of multiple disease classes makes the dataset suitable for multi-class classification using CNN-based deep learning techniques.



3.2 Tools and Technologies

3.2.1 Python

Python is used as the primary programming language for implementing this project. It is widely used in machine learning and deep learning applications due to its simplicity, readability, and extensive library support. Python enables rapid development, experimentation, and integration of data preprocessing, model training, evaluation, and prediction tasks within a single environment.

3.2.2 NumPy

NumPy is a core Python library used for numerical computing. It provides efficient support for multi-dimensional arrays and mathematical operations required for image preprocessing, normalization, and tensor manipulation during CNN training.

3.2.3 Pandas

Pandas is a Python library used for data manipulation and analysis. In this project, Pandas is utilized to manage dataset metadata, organize image labels, analyze class distributions, and handle structured data such as training logs and evaluation results. Its DataFrame structure simplifies data inspection and preprocessing tasks.

3.2.4 Scikit-learn

Scikit-learn is used to support data preprocessing and model evaluation tasks. Although the core classification is performed using a CNN, Scikit-learn utilities are applied for tasks such as accuracy calculation, confusion matrix generation, and performance metric evaluation.

3.2.5 TensorFlow

TensorFlow is an open-source deep learning framework used to build and train the Convolutional Neural Network model. It provides efficient numerical computation using tensor operations and supports scalable execution on both CPU and GPU platforms.

3.2.6 Keras

Keras is a high-level deep learning API built on top of TensorFlow and is used to design and implement the CNN architecture. It enables easy creation of convolutional layers, pooling layers, activation functions, dropout layers, and fully connected layers, making it suitable for image classification tasks.

3.2.7 Jupyter Notebook

Jupyter Notebook is used as the interactive development environment for implementing, training, and testing the CNN model. It supports live code execution, visualization of results, and documentation of experiments, which improves reproducibility and analysis.

3.3 Results

```
Epoch 1/5
/usr/local/lib/python3.12/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDataset`
  self._warn_if_super_not_called()
1358/1358 ————— 98s 67ms/step - accuracy: 0.6011 - loss: 1.6260 - val_accuracy: 0.8659 - val_loss: 0.4379
Epoch 2/5
1358/1358 ————— 94s 69ms/step - accuracy: 0.9167 - loss: 0.2631 - val_accuracy: 0.8541 - val_loss: 0.4824
Epoch 3/5
1358/1358 ————— 87s 64ms/step - accuracy: 0.9673 - loss: 0.1040 - val_accuracy: 0.8916 - val_loss: 0.4068
Epoch 4/5
1358/1358 ————— 143s 65ms/step - accuracy: 0.9804 - loss: 0.0629 - val_accuracy: 0.8860 - val_loss: 0.4814
Epoch 5/5
1358/1358 ————— 87s 64ms/step - accuracy: 0.9825 - loss: 0.0548 - val_accuracy: 0.8795 - val_loss: 0.5401
```

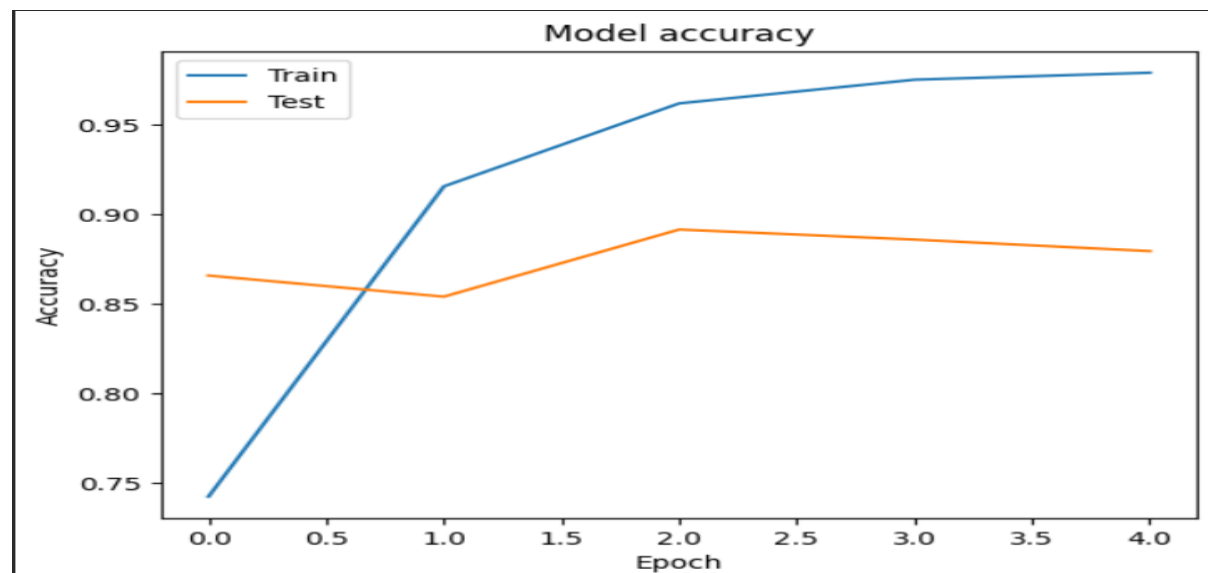
```
print(os.listdir( plantvillage dataset ))

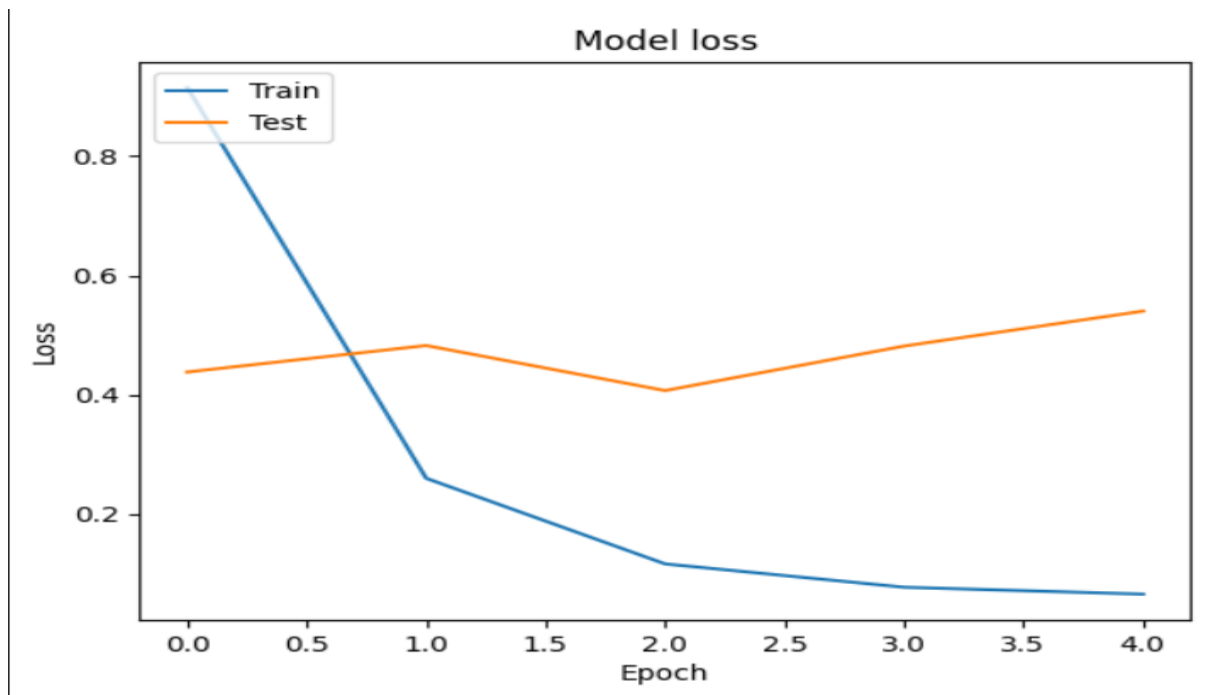
print(len(os.listdir("plantvillage dataset/segmented")))
print(os.listdir("plantvillage dataset/segmented")[:5])

print(len(os.listdir("plantvillage dataset/color")))
print(os.listdir("plantvillage dataset/color")[:5])

print(len(os.listdir("plantvillage dataset/grayscale")))
print(os.listdir("plantvillage dataset/grayscale")[:5])

['color', 'segmented', 'grayscale']
38
['Tomato__healthy', 'Grape__Esca (Black Measles)', 'Corn (maize)__Common_rust_', 'Tomato__Leaf_Mold', 'Grape__healthy']
38
['Tomato__healthy', 'Grape__Esca (Black Measles)', 'Corn (maize)__Common_rust_', 'Tomato__Leaf_Mold', 'Grape__healthy']
38
['Tomato__healthy', 'Grape__Esca (Black Measles)', 'Corn (maize)__Common_rust_', 'Tomato__Leaf_Mold', 'Grape__healthy']
```





Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
flatten (Flatten)	(None, 186624)	0
dense (Dense)	(None, 256)	47,776,000
dense_1 (Dense)	(None, 38)	9,766

Total params: 47,805,158 (182.36 MB)
Trainable params: 47,805,158 (182.36 MB)
Non-trainable params: 0 (0.00 B)


```
# Train Generator
train_generator = data_gen.flow_from_directory(
    base_dir,
    target_size=(img_size, img_size),
    batch_size=batch_size,
    subset='training',
    class_mode='categorical'
)
```

Found 43456 images belonging to 38 classes.

```
# Validation Generator
validation_generator = data_gen.flow_from_directory(
    base_dir,
    target_size=(img_size, img_size),
    batch_size=batch_size,
    subset='validation',
    class_mode='categorical'
)
```

Found 10849 images belonging to 38 classes.

```
# Example Usage
image_path = '/content/test_apple_black_rot.JPG'
#image_path = '/content/test_blueberry_healthy.jpg'
#image_path = '/content/test_potato_early_blight.jpg'
predicted_class_name = predict_image_class(model, image_path, class_indices)

# Output the result
print("Predicted Class Name:", predicted_class_name)
```

```
1/1 ————— 1s 1s/step
Predicted Class Name: Apple__Black_rot
```

Dashboard

Select Page

Home



Welcome to the Plant Disease Recognition System 🌿🔍

This system helps in identifying plant diseases efficiently using **Convolutional Neural Networks (CNN)**.

How It Works

1. Upload a plant leaf image
2. The trained model analyzes the image
3. The disease class is predicted instantly

Dashboard

Select Page

About Dataset



About Dataset

This dataset is recreated using offline augmentation from the original dataset.

- Total 87K RGB images
- 38 classes of healthy and diseased crop leaves
- Split into training and validation sets
- Separate test folder with 33 images

The dataset is used to train a CNN-based plant disease classification model.

Dashboard

Select Page

Prediction

Plant Disease Classifier

Upload an image...



Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG

Browse files



test_blueberry_healthy.jpg 18.5KB



Classify

Prediction: Blueberry__healthy

Chapter 4: Conclusion and Future work

4.1 Conclusion

In this project, a **Convolutional Neural Network (CNN)**–based system for automatic leaf disease detection was successfully designed and implemented. The proposed approach addresses the challenges faced by farmers in identifying plant diseases at an early stage by providing an efficient and accurate image-based diagnostic solution.

A large-scale **Kaggle plant leaf disease dataset** consisting of approximately **87,000 RGB images** across **38 different classes** was used to train and validate the model. The dataset included healthy and diseased leaf images of crops such as **apple, grape, corn, potato, and tomato**. Image preprocessing and data augmentation techniques were applied to enhance data quality and improve model generalization.

The CNN model automatically learned discriminative features from raw leaf images through convolutional, pooling, and fully connected layers, eliminating the need for manual feature extraction. The model was trained using a structured workflow with training and validation datasets, and early stopping was employed to prevent overfitting. Experimental results demonstrated that the proposed CNN model achieved high classification accuracy and performed effectively in identifying multiple leaf disease categories.

The developed system is capable of predicting whether a leaf is healthy or diseased and identifying the corresponding disease type from a given input image. Due to its accuracy, scalability, and automation, the proposed approach can be effectively used in real-world agricultural applications to support farmers and agricultural experts in disease monitoring and crop management.

Overall, this project highlights the potential of deep learning techniques in precision agriculture and demonstrates that CNN-based models are a reliable and practical solution for plant leaf disease detection.